Hongyi Jing

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EDUCATION

University of Southern California (USC), Los Angeles, United States Aug. 2024 – May. 2026
Master of Science, USC Viterbi School of Engineering
Major: Computer Science; **Cumulative GPA:** 3.75/4.0

Huazhong University of Science and Technology (HUST), Wuhan, China Sept. 2020 – Jun. 2024
Bachelor of Engineering, School of Artificial Intelligence and Automation
Major: Artificial Intelligence; **Cumulative GPA:** 3.90/4.0

PUBLICATIONS (* INDICATES EQUAL CONTRIBUTION.)

- Zhenyu Zhao*, **Hongyi Jing***, Xiawei Liu, Jiageng Mao, Abha Jia, Hanwen Yang, Rong Xue, Sergey Zakharor, Vitor Guizilini and Yue Wang. "Humanoid Everyday: A Comprehensive Robotic Dataset for Open-World Humanoid Manipulation", *ICRA 2026.*
- Yuwen Tan, Xiang Xiang, Yifeng Chen, **Hongyi Jing**, Shiyang Ye, Chaoran Xue and Hui Xu. "Coupling Bracket Segmentation and Tooth Surface Reconstruction on 3D Dental Models", *MICCAI 2023*.

RESEARCH EXPERIENCES

Humanoid Manipulation Learning and Evaluation Dec. 2024 – NOW
Research Lead (Advisor: Prof. Yue Wang)
Lab: Physical Superintelligence (PSI) Lab[1], University of Southern California

- Presented Humanoid Everyday, a diverse and multi-modal humanoid manipulation dataset comprising 10.3k trajectories with an optimized teleoperation pipeline.
- Conducted an analysis of the performance of representative policy learning models on Humanoid Everyday, including imitation learning and VLA models, highlighting their strengths and limitations.
- Built a cloud-based evaluation platform that allows researchers to seamlessly deploy their policies in our controlled setting and receive performance feedback.
- Developing an accurate and responsive whole-body humanoid teleoperation pipeline, enabling the training of full-body loco-manipulation VLA policies for long-horizon tasks in complex real-world environments.

Large Language Model in Chemistry Jul. 2023 – Apr. 2024
Research Assistant (Advisor: Prof. Xiangliang Zhang)
Lab: Machine Intelligence and kNowledge Engineering (MINE) Lab[2], University of Notre Dame

- Explored the application of large language models (LLMs) in solving diverse chemical problems, including reaction prediction, relation extraction, and named entity recognition.
- Evaluated the capability boundaries of LLMs in chemical information extraction tasks, comparing their performance against specialized models to analyze the advantages and shortcomings of LLMs in chemical tasks.
- Revisited and improved the existing evaluation metrics for LLMs in scientific information extraction, and establish a comprehensive system for evaluating the chemical capabilities of LLMs.

3D Dental Model Segmentation and Reconstruction Aug. 2022 – Mar. 2023
Research Assistant (Advisor: Prof. Xiang Xiang)
Lab: Human-AI Interaction and Visualization (HAIV) Lab[3], HUST

- Employed deep learning networks and projection-based surface reconstruction to develop an efficient approach for tooth segmentation and 3D reconstruction in orthodontic treatment.
- Designed a graph-based network for bracket segmentation on 3D dental models, and performed post-processing after bracket segmentation to optimize tooth edges to remove inaccuracies and enhance surface smoothness.
- Implemented 3D dental model reconstruction after bracket removal by projecting the dental surface onto a 2D plane, repairing it based on curvature information, and re-projecting it onto the original 3D tooth surface.

ADDITIONAL INFORMATION

- **Programming:** Python, C/C++, C#, Matlab.
- **Developer Tools:** Git, Docker, VS Code, Visual Studio, PyCharm, Unity.
- **Research Interests:** Robotics, Dexterous Manipulation, Whole-Body Control, Vision-Language Models.

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[1] Official website of GVL Lab is available at: https://psi-lab.ai/lab.html
[2] Official website of MINE Lab is available at: https://sites.nd.edu/xiangliang-zhang/
[3] Official website of HAIV Lab is available at: https://haivlab.wixsite.com/home